

Madrid Air Quality

A reproducible analytics pipeline on the METRAQ dataset (2001–2024)

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Problem & Goal

What we set out to answer with the METRAQ dataset

Air quality is a public-health concern. Cities need evidence-based tools to monitor and reduce harm, and 24 years of hourly observations let us look hard at the question.



WHERE

What are the current pollution patterns in Madrid?



WHEN

At what times of the day or year do pollution levels reach their peak?



HOW

Which areas are most affected, and which sensors behave alike?



WHY

How do pollutants relate to weather and traffic?



Dataset Inspection

METRAQ Air Quality, hourly observations 2001-2024

~65M

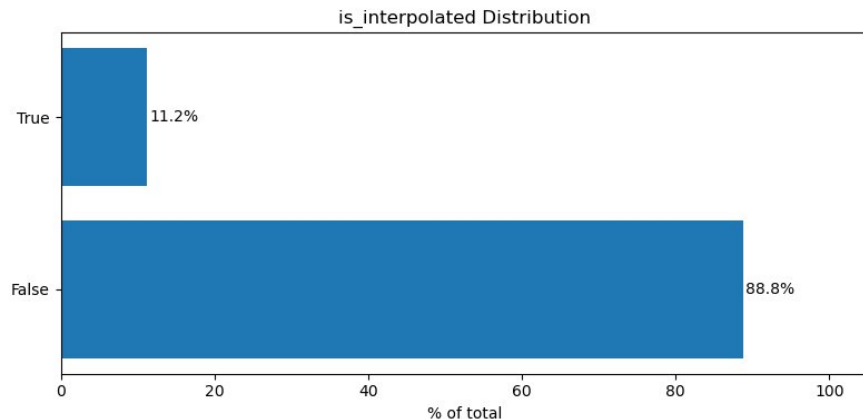
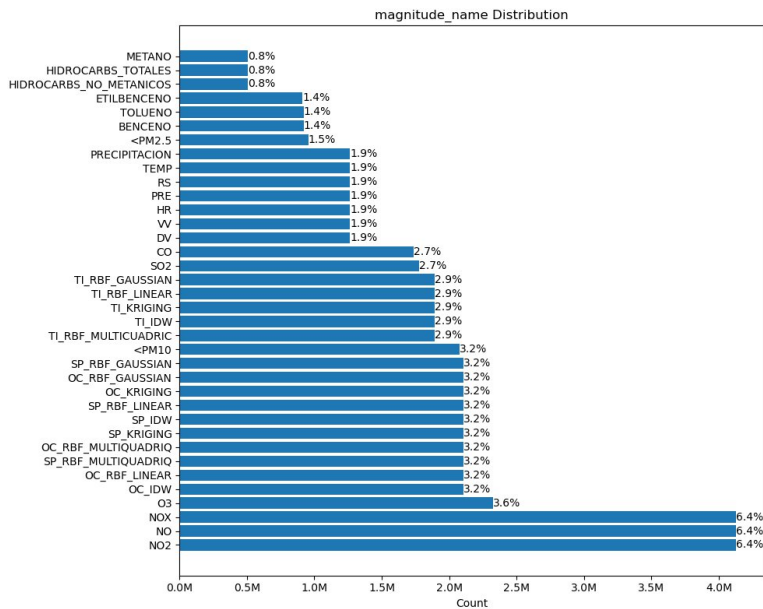
rows of hourly data

24

years of coverage

36

measured variables (14 pollutants + 7 weather + 15 traffic)

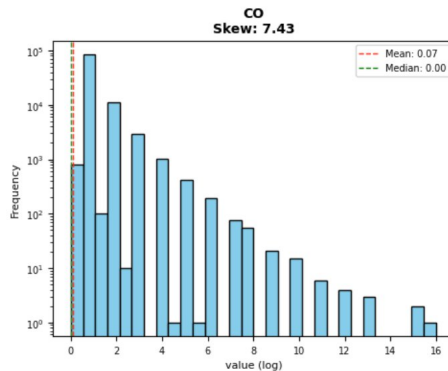
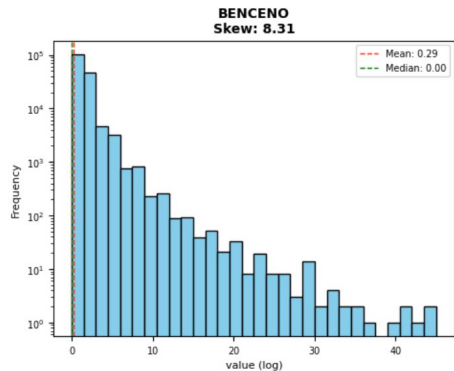
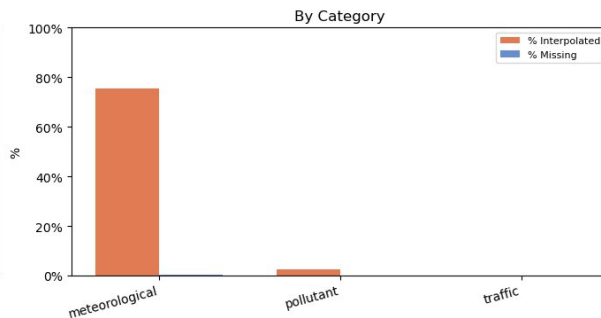
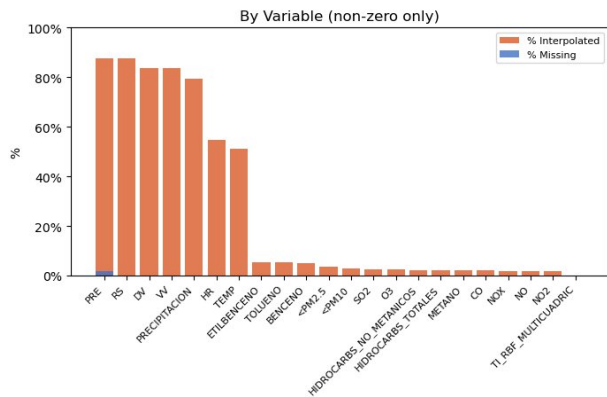




Missingness & data quality

METRAQ already imputed missingness — we kept the flag and re-evaluated

Missingness & Interpolation Summary



Outlier philosophy

NaN only physically impossible values.

Keep heavy-tailed pollutants intact.

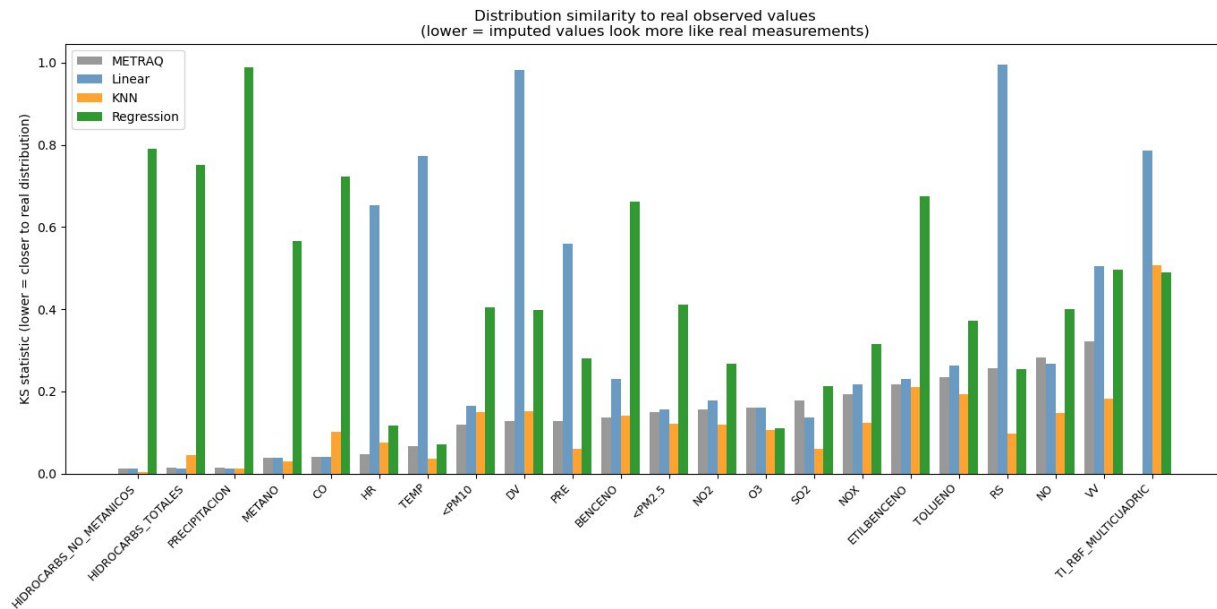
Why?

Benceno & CO are zero-inflated by design, sensors read 0 most of the time and spike during episodes. Treating those tails as outliers would delete the signal.



Imputation: four methods, one ranking

KS distance to the real distribution — lower is better



Methods compared

METRAQ spatial interpolation(baseline)

Linear temporal interpolation

Regression global linear model

KNN k=2, distance-weighted

12 / 22 Variables where our methods beat METRAQ

06 / 22 Ties within 0.01 KS

04 / 22 Variables where METRAQ wins



Best-of-breed dataset construction

Pick the per-variable winner

```
=== Best-of-breed strategy per variable ===
              chosen_method  KNN  Linear  METRAQ  Regression
NO              KNN    0.148  0.268  0.282    0.401
TEMP            KNN    0.039  0.761  0.065    0.071
SO2             KNN    0.060  0.136  0.177    0.213
RS              KNN    0.098  0.996  0.256    0.253
PRE             KNN    0.060  0.559  0.128    0.281
O3              KNN    0.106  0.160  0.161    0.110
NOX             KNN    0.124  0.216  0.192    0.314
NO2             KNN    0.119  0.178  0.157    0.268
TOLUENO         KNN    0.194  0.263  0.234    0.372
VV              KNN    0.182  0.505  0.322    0.495
<PM2.5          KNN    0.120  0.157  0.150    0.412
HR              METRAQ  0.075  0.653  0.048    0.117
HIDROCARBS_TOTALES METRAQ  0.044  0.013  0.014    0.750
HIDROCARBS_NO_METANICOS METRAQ  0.003  0.012  0.011    0.790
ETILBENCENO     METRAQ  0.210  0.229  0.217    0.675
DV              METRAQ  0.151  0.983  0.127    0.397
PRECIPITACION   METRAQ  0.012  0.012  0.014    0.988
CO              METRAQ  0.101  0.041  0.041    0.722
BENCENO         METRAQ  0.141  0.231  0.136    0.661
METANO          METRAQ  0.030  0.038  0.039    0.566
<PM10           METRAQ  0.150  0.164  0.119    0.404
TI_RBF_MULTICUADRIC Regression 0.507  0.786  NaN     0.490
```

```
Method counts: {'KNN': 11, 'METRAQ': 10, 'Regression': 1}
```

```
Imputed rows written: 7,287,645
```

```
Remaining NaNs in value: 0
```

```
df shape: (64881731, 10)
```

Three steps

1. Pick lowest KS per variable
2. METRAQ wins ties within 0.01
3. KNN baseline + winner override

Final assignment

KNN 11 METRAQ 10 Reg 1

0

NaNs after imputation

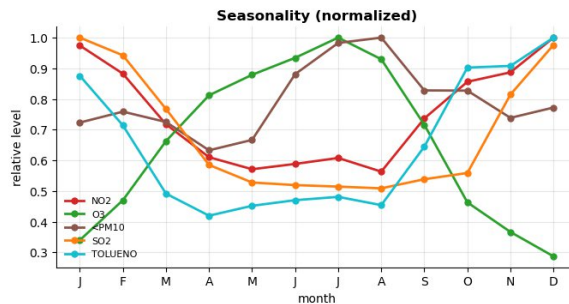
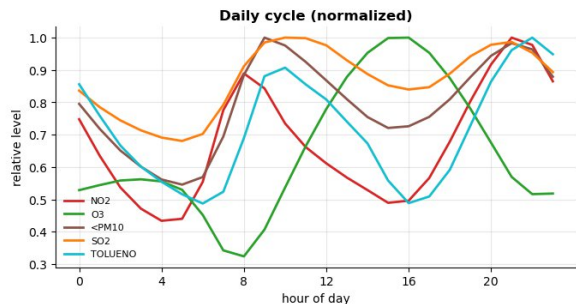
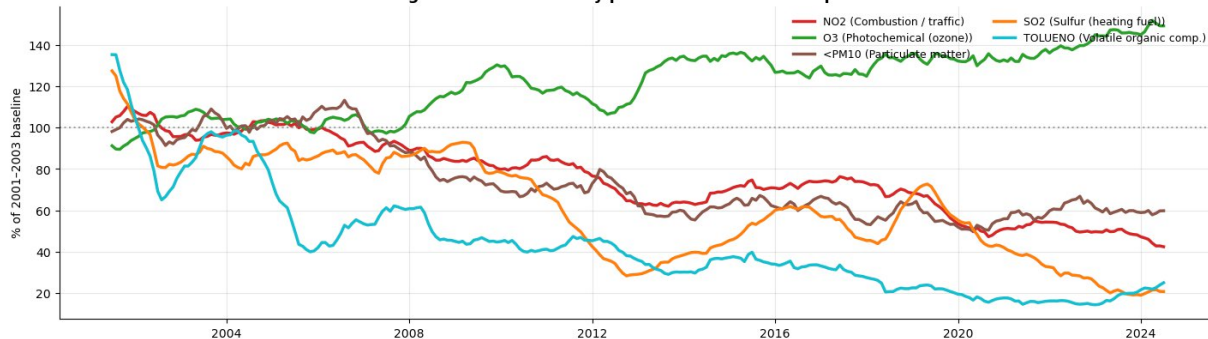
zero rows dropped



Temporal patterns

Monthly aggregation reveals the long-run trend

Long-term trends — every pollutant has fallen except ozone



Findings

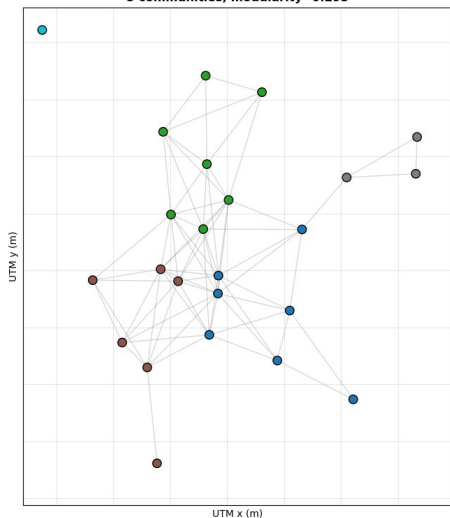
- **NO₂** has dropped substantially over 24 years
- **O₃** has slowly risen (titration effect)
- **Traffic-linked** pollutants show strong daily + yearly cycles
- **Weather** shows clean seasonal cycles, no trend



Spatial sensor network

Two construction methods compared: distance threshold vs. k-NN

Threshold (5 km) — communities
5 communities, modularity=0.293



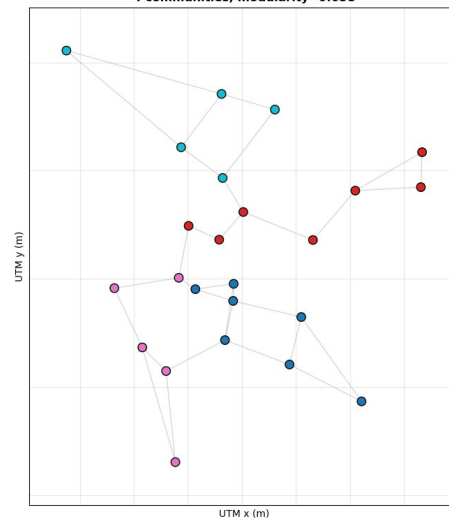
Distance threshold:

Threshold	Edges	Avg degree	Components	Isolated	Modularity
1.5 km	2	0.17	22	20	0.497
3 km	26	2.17	11	6	0.372
5 km	73	6.08	2	1	0.293
8 km	148	12.33	1	0	0.104
15 km	259	21.58	1	0	0.000

KNN:

k	Edges	Avg degree	Components	Isolated	Modularity
1	16	1.33	8	0	0.801
2	32	2.67	1	0	0.658
3	50	4.17	1	0	0.538
5	84	7.00	1	0	0.396
8	130	10.83	1	0	0.300

KNN (k=2) — communities
4 communities, modularity=0.658



Distance threshold

Ties to physical reality, but density-sensitive — full connectivity at low thresholds.

k-Nearest Neighbours

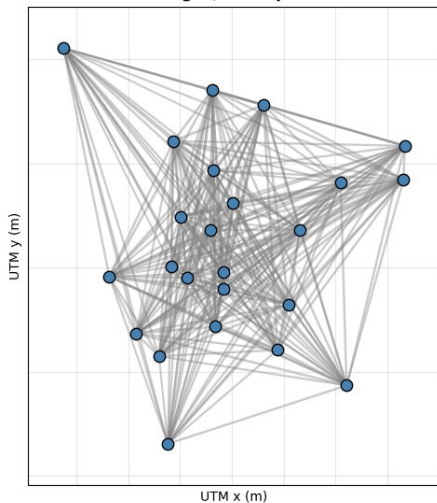
Stable degree distribution, but ignores absolute distance between sensors.



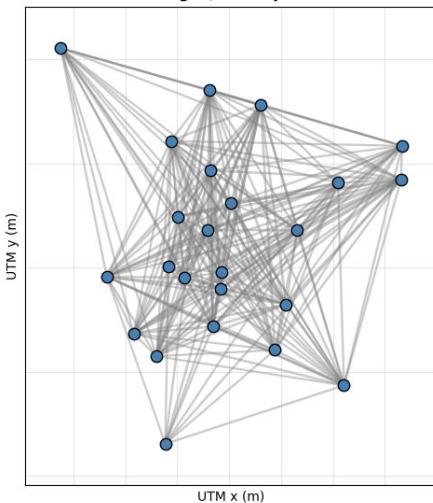
Correlation network

Sensors connected by similar behaviour over time, not proximity

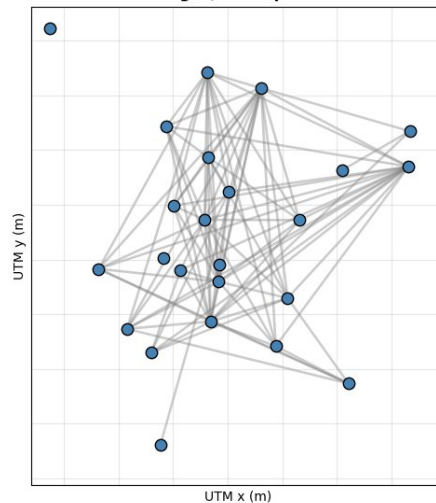
NO2 correlation graph (threshold=0.7)
272 edges, 1 components



NO2 correlation graph (threshold=0.8)
232 edges, 1 components



NO2 correlation graph (threshold=0.9)
89 edges, 4 components



Method

- Monthly aggregation per sensor
- Pairwise Pearson correlation
- Edge if $|\text{corr}| > \text{threshold}$

Insight

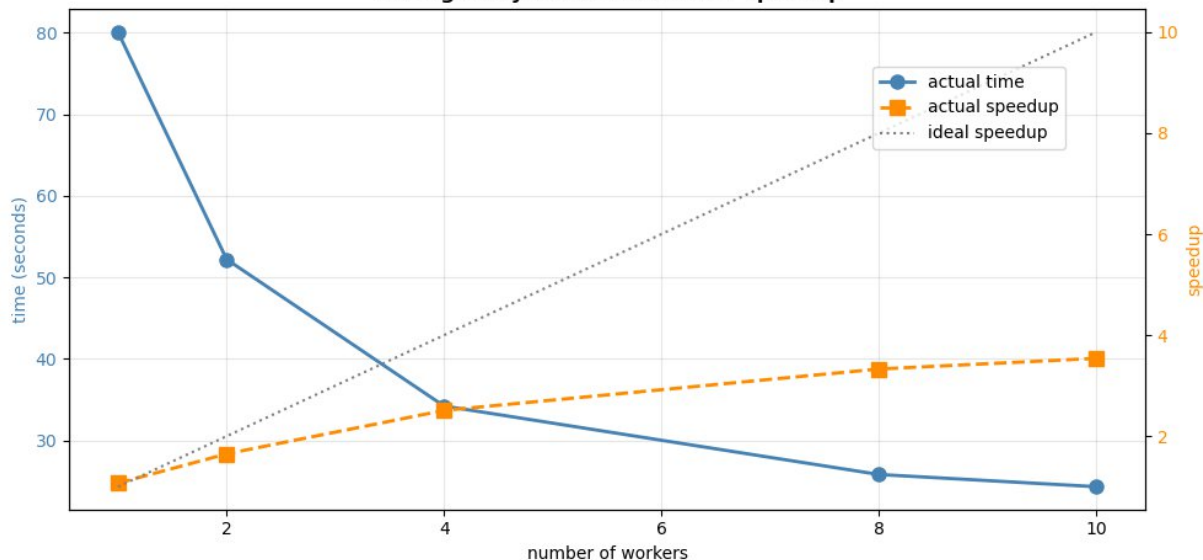
Correlation network / distance network. Sensors far apart can behave alike, shared wind regime, shared traffic rhythm.



Parallel correlations: scaling study

Hourly correlation matrix per (year, sensor) — sequential vs. multiprocessing.Pool

Scaling analysis: actual vs ideal speedup



Setup

Workers: **1, 2, 4, 8, 10**

Each worker reads its own per-year parquet

Result

Near-linear speedup

up to ~4 workers (78% efficiency at 2, 59% at 4)

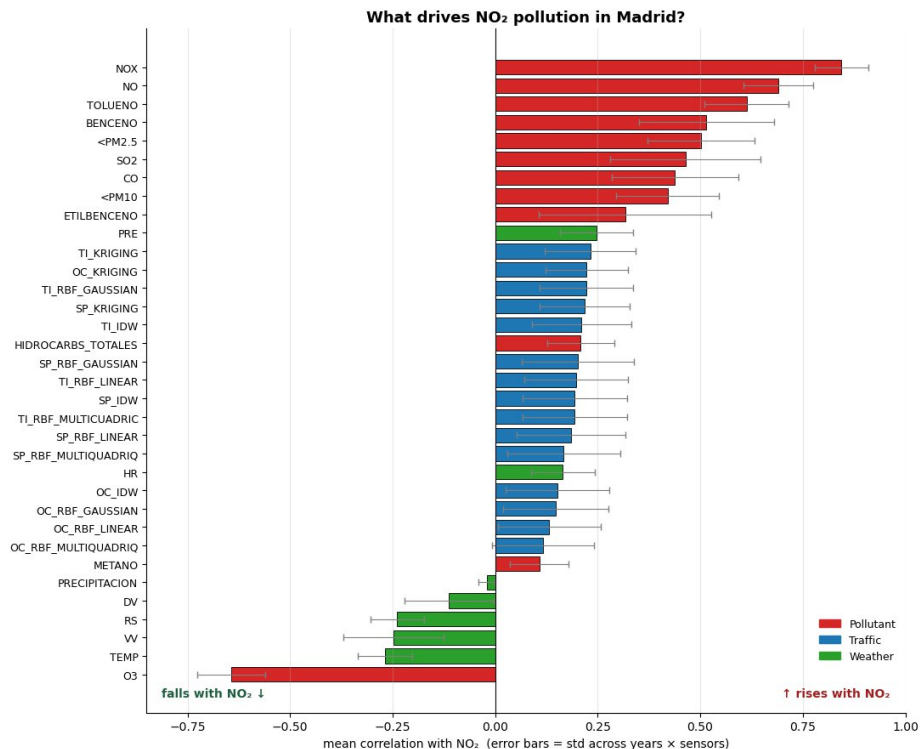
Diminishing returns beyond 4 workers

process overhead dominates, each task ~150 ms, so forking costs matter. Optimal: 8 workers (3.26x, 41% eff.)



Which variables move with NO₂?

Median + mean across all (year, sensor) cells



TOP POSITIVE

NOX
NO
Traffic intensity

TOP NEGATIVE

Wind speed
Ozone



Forecasting NO₂ from confounders alone

Random Forest vs linear baseline — 2023 held-out test set

R A N D O M F O R E S T

$R^2 = 0.76$

RMSE = 3.7 $\mu\text{g}/\text{m}^3$

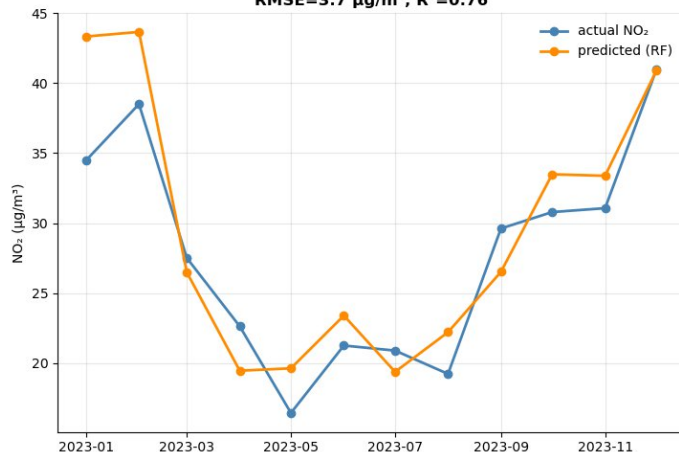
MAE = 3.0 $\mu\text{g}/\text{m}^3$

L I N E A R B A S E L I N E

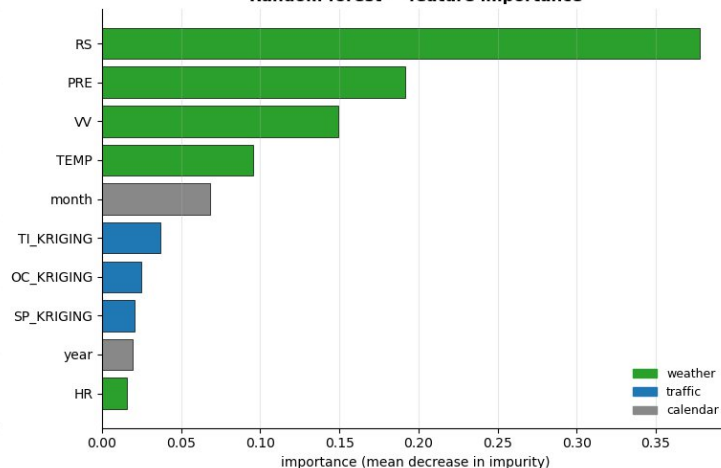
$R^2 = -1.22$

RMSE = 11.2 $\mu\text{g}/\text{m}^3$

Test set 2023 — monthly means
RMSE=3.7 $\mu\text{g}/\text{m}^3$, $R^2=0.76$



Random forest — feature importance





Synthesis: four stories of Madrid air

WHERE · WHEN · HOW · WHY



W H E R E

Sensor Map

NO₂ hotspots concentrate in central traffic corridors.



W H E N

24 Year Trend

NO₂ declines, O₃ rises, the photochemistry trade-off.



H O W

Correlation Network

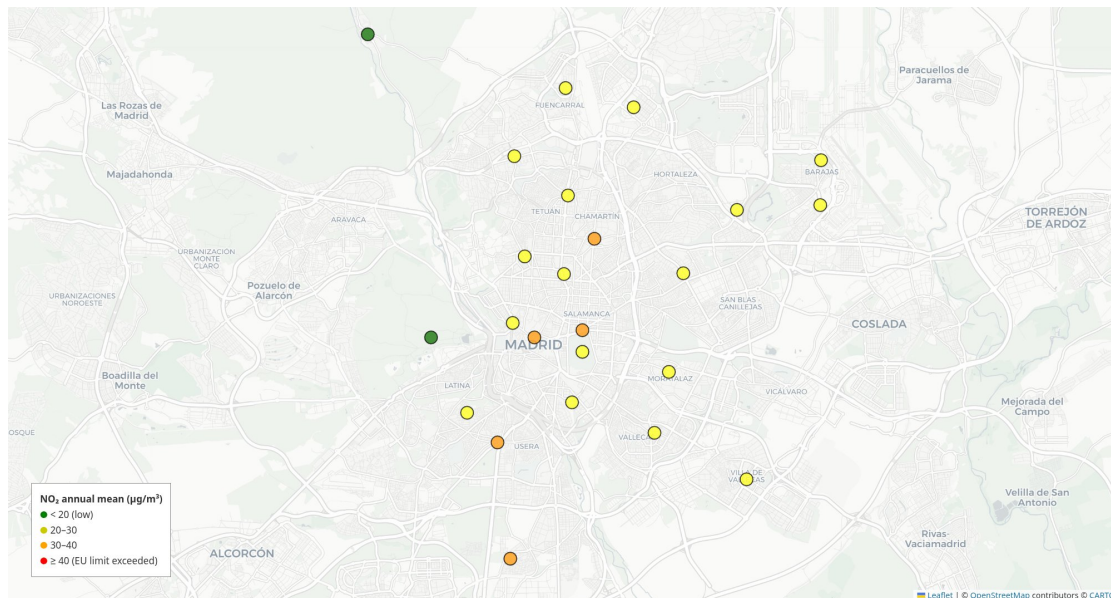
Sensors far apart still behave alike when wind & traffic align.



W H Y

Wind vs. Traffic

NO₂ negatively correlates with wind speed; positively with traffic.





Thank you

Questions?

Code & full pipeline reproducible from [GitHub](#)

Dataset: [METRAQ Air Quality \(Hugging Face\)](#)

[Group Matteos]